

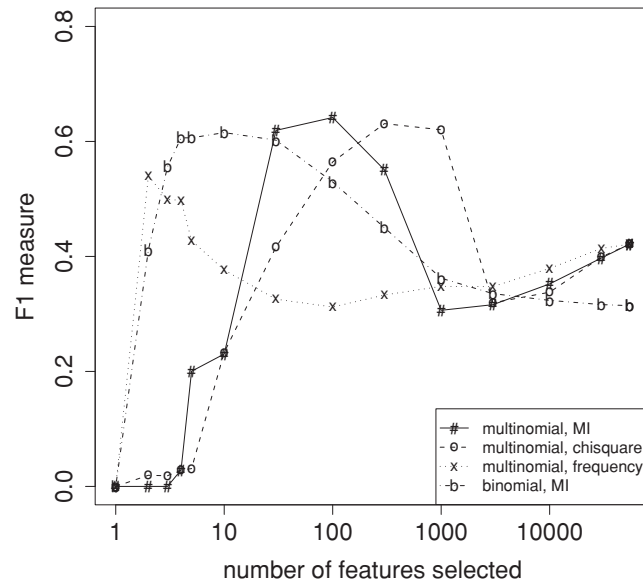
<i>UK</i>		<i>China</i>		<i>poultry</i>	
london	0.1925	china	0.0997	poultry	0.0013
uk	0.0755	chinese	0.0523	meat	0.0008
british	0.0596	beijing	0.0444	chicken	0.0006
stg	0.0555	yuan	0.0344	agriculture	0.0005
britain	0.0469	shanghai	0.0292	avian	0.0004
plc	0.0357	hong	0.0198	broiler	0.0003
england	0.0238	kong	0.0195	veterinary	0.0003
pence	0.0212	xinhua	0.0155	birds	0.0003
pounds	0.0149	province	0.0117	inspection	0.0003
english	0.0126	taiwan	0.0108	pathogenic	0.0003
<i>coffee</i>		<i>elections</i>		<i>sports</i>	
coffee	0.0111	election	0.0519	soccer	0.0681
bags	0.0042	elections	0.0342	cup	0.0515
growers	0.0025	polls	0.0339	match	0.0441
kg	0.0019	voters	0.0315	matches	0.0408
colombia	0.0018	party	0.0303	played	0.0388
brazil	0.0016	vote	0.0299	league	0.0386
export	0.0014	poll	0.0225	beat	0.0301
exporters	0.0013	candidate	0.0202	game	0.0299
exports	0.0013	campaign	0.0202	games	0.0284
crop	0.0012	democratic	0.0198	team	0.0264

► **Figure 13.7** Features with high mutual information scores for six Reuters-RCV1 classes.

whether the document is in the class. As you might expect, keeping the informative terms and eliminating the non-informative ones tends to reduce noise and improve the classifier's accuracy.

Such an accuracy increase can be observed in Figure 13.8, which shows F_1 as a function of vocabulary size after feature selection for Reuters-RCV1.⁷ Comparing F_1 at 132,776 features (corresponding to selection of all features) and at 10–100 features, we see that MI feature selection increases F_1 by about 0.1 for the multinomial model and by more than 0.2 for the Bernoulli model. For the Bernoulli model, F_1 peaks early, at ten features selected. At that point, the Bernoulli model is better than the multinomial model. When basing a classification decision on only a few features, it is more robust to consider binary occurrence only. For the multinomial model (MI feature selection), the peak occurs later, at 100 features, and its effectiveness recovers somewhat at

7. We trained the classifiers on the first 100,000 documents and computed F_1 on the next 100,000. The graphs are averages over five classes.



► **Figure 13.8** Effect of feature set size on accuracy for multinomial and Bernoulli models.

the end when we use all features. The reason is that the multinomial takes the number of occurrences into account in parameter estimation and classification and therefore better exploits a larger number of features than the Bernoulli model. Regardless of the differences between the two methods, using a carefully selected subset of the features results in better effectiveness than using all features.

13.5.2 χ^2 Feature selection

χ^2 FEATURE SELECTION

INDEPENDENCE

Another popular feature selection method is χ^2 . In statistics, the χ^2 test is applied to test the independence of two events, where two events A and B are defined to be *independent* if $P(AB) = P(A)P(B)$ or, equivalently, $P(A|B) = P(A)$ and $P(B|A) = P(B)$. In feature selection, the two events are occurrence of the term and occurrence of the class. We then rank terms with respect to the following quantity:

$$(13.18) \quad X^2(\mathbb{D}, t, c) = \sum_{e_t \in \{0,1\}} \sum_{e_c \in \{0,1\}} \frac{(N_{e_t e_c} - E_{e_t e_c})^2}{E_{e_t e_c}}$$

where e_t and e_c are defined as in Equation (13.16). N is the *observed* frequency in $\mathbb{D}$ and E the *expected* frequency. For example, E_{11} is the expected frequency of t and c occurring together in a document assuming that term and class are independent.



Example 13.4: We first compute E_{11} for the data in Example 13.3:

$$\begin{aligned} E_{11} &= N \times P(t) \times P(c) = N \times \frac{N_{11} + N_{10}}{N} \times \frac{N_{11} + N_{01}}{N} \\ &= N \times \frac{49 + 141}{N} \times \frac{49 + 27652}{N} \approx 6.6 \end{aligned}$$

where N is the total number of documents as before.

We compute the other $E_{e_t e_c}$ in the same way:

	$e_{poultry} = 1$		$e_{poultry} = 0$	
$e_{\text{export}} = 1$	$N_{11} = 49$	$E_{11} \approx 6.6$	$N_{10} = 27,652$	$E_{10} \approx 27,694.4$
$e_{\text{export}} = 0$	$N_{01} = 141$	$E_{01} \approx 183.4$	$N_{00} = 774,106$	$E_{00} \approx 774,063.6$

Plugging these values into Equation (13.18), we get a X^2 value of 284:

$$X^2(\mathbb{D}, t, c) = \sum_{e_t \in \{0,1\}} \sum_{e_c \in \{0,1\}} \frac{(N_{e_t e_c} - E_{e_t e_c})^2}{E_{e_t e_c}} \approx 284$$

X^2 is a measure of how much expected counts E and observed counts N deviate from each other. A high value of X^2 indicates that the hypothesis of independence, which implies that expected and observed counts are similar, is incorrect. In our example, $X^2 \approx 284 > 10.83$. Based on Table 13.6, we can reject the hypothesis that *poultry* and *export* are independent with only a 0.001 chance of being wrong.⁸ Equivalently, we say that the outcome $X^2 \approx 284 > 10.83$ is *statistically significant* at the 0.001 level. If the two events are dependent, then the occurrence of the term makes the occurrence of the class more likely (or less likely), so it should be helpful as a feature. This is the rationale of χ^2 feature selection.

An arithmetically simpler way of computing X^2 is the following:

$$(13.19) \quad X^2(\mathbb{D}, t, c) = \frac{(N_{11} + N_{10} + N_{01} + N_{00}) \times (N_{11}N_{00} - N_{10}N_{01})^2}{(N_{11} + N_{01}) \times (N_{11} + N_{10}) \times (N_{10} + N_{00}) \times (N_{01} + N_{00})}$$

This is equivalent to Equation (13.18) (Exercise 13.14).

8. We can make this inference because, if the two events are independent, then $X^2 \sim \chi^2$, where χ^2 is the χ^2 distribution. See, for example, Rice (2006).

► **Table 13.6** Critical values of the χ^2 distribution with one degree of freedom. For example, if the two events are independent, then $P(X^2 > 6.63) < 0.01$. So for $X^2 > 6.63$ the assumption of independence can be rejected with 99% confidence.

p	χ^2 critical value
0.1	2.71
0.05	3.84
0.01	6.63
0.005	7.88
0.001	10.83



Assessing χ^2 as a feature selection method

From a statistical point of view, χ^2 feature selection is problematic. For a test with one degree of freedom, the so-called Yates correction should be used (see Section 13.7), which makes it harder to reach statistical significance. Also, whenever a statistical test is used multiple times, then the probability of getting at least one error increases. If 1,000 hypotheses are rejected, each with 0.05 error probability, then $0.05 \times 1000 = 50$ calls of the test will be wrong on average. However, in text classification it rarely matters whether a few additional terms are added to the feature set or removed from it. Rather, the *relative* importance of features is important. As long as χ^2 feature selection only ranks features with respect to their usefulness and is not used to make statements about statistical dependence or independence of variables, we need not be overly concerned that it does not adhere strictly to statistical theory.

13.5.3 Frequency-based feature selection

A third feature selection method is *frequency-based feature selection*, that is, selecting the terms that are most common in the class. Frequency can be either defined as document frequency (the number of documents in the class c that contain the term t) or as collection frequency (the number of tokens of t that occur in documents in c). Document frequency is more appropriate for the Bernoulli model, collection frequency for the multinomial model.

Frequency-based feature selection selects some frequent terms that have no specific information about the class, for example, the days of the week (Monday, Tuesday, ...), which are frequent across classes in newswire text. When many thousands of features are selected, then frequency-based feature selection often does well. Thus, if somewhat suboptimal accuracy is acceptable, then frequency-based feature selection can be a good alternative to more complex methods. However, Figure 13.8 is a case where frequency-

based feature selection performs a lot worse than MI and χ^2 and should not be used.

13.5.4 Feature selection for multiple classifiers

In an operational system with a large number of classifiers, it is desirable to select a single set of features instead of a different one for each classifier. One way of doing this is to compute the X^2 statistic for an $n \times 2$ table where the columns are occurrence and nonoccurrence of the term and each row corresponds to one of the classes. We can then select the k terms with the highest X^2 statistic as before.

More commonly, feature selection statistics are first computed separately for each class on the two-class classification task c versus $\bar{c}$ and then combined. One combination method computes a single figure of merit for each feature, for example, by averaging the values $A(t, c)$ for feature t , and then selects the k features with highest figures of merit. Another frequently used combination method selects the top k/n features for each of n classifiers and then combines these n sets into one global feature set.

Classification accuracy often decreases when selecting k common features for a system with n classifiers as opposed to n different sets of size k . But even if it does, the gain in efficiency owing to a common document representation may be worth the loss in accuracy.

13.5.5 Comparison of feature selection methods

Mutual information and χ^2 represent rather different feature selection methods. The independence of term t and class c can sometimes be rejected with high confidence even if t carries little information about membership of a document in c . This is particularly true for rare terms. If a term occurs once in a large collection and that one occurrence is in the *poultry* class, then this is statistically significant. But a single occurrence is not very informative according to the information-theoretic definition of information. Because its criterion is significance, χ^2 selects more rare terms (which are often less reliable indicators) than mutual information. But the selection criterion of mutual information also does not necessarily select the terms that maximize classification accuracy.

Despite the differences between the two methods, the classification accuracy of feature sets selected with χ^2 and MI does not seem to differ systematically. In most text classification problems, there are a few strong indicators and many weak indicators. As long as all strong indicators and a large number of weak indicators are selected, accuracy is expected to be good. Both methods do this.

Figure 13.8 compares MI and χ^2 feature selection for the multinomial model.

Peak effectiveness is virtually the same for both methods. χ^2 reaches this peak later, at 300 features, probably because the rare, but highly significant features it selects initially do not cover all documents in the class. However, features selected later (in the range of 100–300) are of better quality than those selected by MI.

GREEDY FEATURE
SELECTION

All three methods – MI, χ^2 and frequency based – are *greedy* methods. They may select features that contribute no incremental information over previously selected features. In Figure 13.7, kong is selected as the seventh term even though it is highly correlated with previously selected hong and therefore redundant. Although such redundancy can negatively impact accuracy, non-greedy methods (see Section 13.7 for references) are rarely used in text classification due to their computational cost.

?

Exercise 13.5

Consider the following frequencies for the class *coffee* for four terms in the first 100,000 documents of Reuters-RCV1:

term	N_{00}	N_{01}	N_{10}	N_{11}
brazil	98,012	102	1835	51
council	96,322	133	3525	20
producers	98,524	119	1118	34
roasted	99,824	143	23	10

Select two of these four terms based on (i) χ^2 , (ii) mutual information, (iii) frequency.

13.6 Evaluation of text classification

] Historically, the classic Reuters-21578 collection was the main benchmark for text classification evaluation. This is a collection of 21,578 newswire articles, originally collected and labeled by Carnegie Group, Inc. and Reuters, Ltd. in the course of developing the CONSTRUE text classification system. It is much smaller than and predates the Reuters-RCV1 collection discussed in Chapter 4 (page 69). The articles are assigned classes from a set of 118 topic categories. A document may be assigned several classes or none, but the commonest case is single assignment (documents with at least one class received an average of 1.24 classes). The standard approach to this *any-of* problem (Chapter 14, page 306) is to learn 118 two-class classifiers, one for each class, where the *two-class classifier* for class c is the classifier for the two classes c and its complement $\bar{c}$.

TWO-CLASS CLASSIFIER

MODAPTE SPLIT

For each of these classifiers, we can measure recall, precision, and accuracy. In recent work, people almost invariably use the *ModApte split*, which includes only documents that were viewed and assessed by a human indexer,

► **Table 13.7** The ten largest classes in the Reuters-21578 collection with number of documents in training and test sets.

class	# train	# test	class	# train	# test
<i>earn</i>	2877	1087	<i>trade</i>	369	119
<i>acquisitions</i>	1650	179	<i>interest</i>	347	131
<i>money-fx</i>	538	179	<i>ship</i>	197	89
<i>grain</i>	433	149	<i>wheat</i>	212	71
<i>crude</i>	389	189	<i>corn</i>	182	56

and comprises 9,603 training documents and 3,299 test documents. The distribution of documents in classes is very uneven, and some work evaluates systems on only documents in the ten largest classes. They are listed in Table 13.7. A typical document with topics is shown in Figure 13.9.

In Section 13.1, we stated as our goal in text classification the minimization of classification error on test data. Classification error is 1.0 minus classification accuracy, the proportion of correct decisions, a measure we introduced in Section 8.3 (page 155). This measure is appropriate if the percentage of documents in the class is high, perhaps 10% to 20% and higher. But as we discussed in Section 8.3, accuracy is not a good measure for “small” classes because always saying no, a strategy that defeats the purpose of building a classifier, will achieve high accuracy. The always-no classifier is 99% accurate for a class with relative frequency 1%. For small classes, precision, recall and F_1 are better measures.

EFFECTIVENESS

PERFORMANCE
EFFICIENCY

We will use *effectiveness* as a generic term for measures that evaluate the quality of classification decisions, including precision, recall, F_1 , and accuracy. *Performance* refers to the computational efficiency of classification and IR systems in this book. However, many researchers mean effectiveness, not efficiency of text classification when they use the term performance.

MACROAVERAGING
MICROAVERAGING

When we process a collection with several two-class classifiers (such as Reuters-21578 with its 118 classes), we often want to compute a single aggregate measure that combines the measures for individual classifiers. There are two methods for doing this. *Macroaveraging* computes a simple average over classes. *Microaveraging* pools per-document decisions across classes, and then computes an effectiveness measure on the pooled contingency table. Table 13.8 gives an example.

The differences between the two methods can be large. Macroaveraging gives equal weight to each class, whereas microaveraging gives equal weight to each per-document classification decision. Because the F_1 measure ignores true negatives and its magnitude is mostly determined by the number of true positives, large classes dominate small classes in microaveraging. In the example, microaveraged precision (0.83) is much closer to the precision of

```

<REUTERS TOPICS=' 'YES' ' LEWISSPLIT=' 'TRAIN' '
CGISPLIT=' 'TRAINING-SET' ' OLDID=' '12981' ' NEWID=' '798' '>
<DATE> 2-MAR-1987 16:51:43.42</DATE>
<TOPICS><D>livestock</D><D>hog</D></TOPICS>
<TITLE>AMERICAN PORK CONGRESS KICKS OFF TOMORROW</TITLE>
<DATELINE> CHICAGO, March 2 - </DATELINE><BODY>The American Pork
Congress kicks off tomorrow, March 3, in Indianapolis with 160
of the nations pork producers from 44 member states determining
industry positions on a number of issues, according to the
National Pork Producers Council, NPPC.
Delegates to the three day Congress will be considering 26
resolutions concerning various issues, including the future
direction of farm policy and the tax law as it applies to the
agriculture sector. The delegates will also debate whether to
endorse concepts of a national PRV (pseudorabies virus) control
and eradication program, the NPPC said. A large
trade show, in conjunction with the congress, will feature
the latest in technology in all areas of the industry, the NPPC
added. Reuter
\&\#3;</BODY></TEXT></REUTERS>

```

► **Figure 13.9** A sample document from the Reuters-21578 collection.

c_2 (0.9) than to the precision of c_1 (0.5) because c_2 is five times larger than c_1 . Microaveraged results are therefore really a measure of effectiveness on the large classes in a test collection. To get a sense of effectiveness on small classes, you should compute macroaveraged results.

In one-of classification (Section 14.5, page 306), microaveraged F_1 is the same as accuracy (Exercise 13.6).

Table 13.9 gives microaveraged and macroaveraged effectiveness of Naive Bayes for the ModApte split of Reuters-21578. To give a sense of the relative effectiveness of NB, we compare it with linear SVMs (rightmost column; see Chapter 15), one of the most effective classifiers, but also one that is more expensive to train than NB. NB has a microaveraged F_1 of 80%, which is 9% less than the SVM (89%), a 10% relative decrease (row “micro-avg-L (90 classes)”). So there is a surprisingly small effectiveness penalty for its simplicity and efficiency. However, on small classes, some of which only have on the order of ten positive examples in the training set, NB does much worse. Its macroaveraged F_1 is 13% below the SVM, a 22% relative decrease (row “macro-avg (90 classes)”).

The table also compares NB with the other classifiers we cover in this book:

► **Table 13.8** Macro- and microaveraging. “Truth” is the true class and “call” the decision of the classifier. In this example, macroaveraged precision is $[10/(10+10) + 90/(10+90)]/2 = (0.5 + 0.9)/2 = 0.7$. Microaveraged precision is $100/(100+20) \approx 0.83$.

	class 1			class 2			pooled table	
	truth: yes	truth: no		truth: yes	truth: no		truth: yes	truth: no
call: yes	10	10	call: yes	90	10	call: yes	100	20
call: no	10	970	call: no	10	890	call: no	20	1860

► **Table 13.9** Text classification effectiveness numbers on Reuters-21578 for F_1 (in percent). Results from [Li and Yang \(2003\)](#) (a), [Joachims \(1998\)](#) (b: kNN) and [Dumais et al. \(1998\)](#) (b: NB, Rocchio, trees, SVM).

(a)	NB	Rocchio	kNN	SVM
micro-avg-L (90 classes)	80	85	86	89
macro-avg (90 classes)	47	59	60	60

(b)	NB	Rocchio	kNN	trees	SVM
earn	96	93	97	98	98
acq	88	65	92	90	94
money-fx	57	47	78	66	75
grain	79	68	82	85	95
crude	80	70	86	85	89
trade	64	65	77	73	76
interest	65	63	74	67	78
ship	85	49	79	74	86
wheat	70	69	77	93	92
corn	65	48	78	92	90
micro-avg (top 10)	82	65	82	88	92
micro-avg-D (118 classes)	75	62	n/a	n/a	87

DECISION TREES

Rocchio and kNN. In addition, we give numbers for *decision trees*, an important classification method we do not cover. The bottom part of the table shows that there is considerable variation from class to class. For instance, NB beats kNN on *ship*, but is much worse on *money-fx*.

Comparing parts (a) and (b) of the table, one is struck by the degree to which the cited papers’ results differ. This is partly due to the fact that the numbers in (b) are break-even scores (cf. page 161) averaged over 118 classes, whereas the numbers in (a) are true F_1 scores (computed without any know-

ledge of the test set) averaged over ninety classes. This is unfortunately typical of what happens when comparing different results in text classification: There are often differences in the experimental setup or the evaluation that complicate the interpretation of the results.

These and other results have shown that the average effectiveness of NB is uncompetitive with classifiers like SVMs when trained and tested on *independent and identically distributed* (i.i.d.) data, that is, uniform data with all the good properties of statistical sampling. However, these differences may often be invisible or even reverse themselves when working in the real world where, usually, the training sample is drawn from a subset of the data to which the classifier will be applied, the nature of the data drifts over time rather than being stationary (the problem of concept drift we mentioned on page 269), and there may well be errors in the data (among other problems). Many practitioners have had the experience of being unable to build a fancy classifier for a certain problem that consistently performs better than NB.

Our conclusion from the results in Table 13.9 is that, although most researchers believe that an SVM is better than kNN and kNN better than NB, the ranking of classifiers ultimately depends on the class, the document collection, and the experimental setup. In text classification, there is always more to know than simply which machine learning algorithm was used, as we further discuss in Section 15.3 (page 334).

When performing evaluations like the one in Table 13.9, it is important to maintain a strict separation between the training set and the test set. We can easily make correct classification decisions on the test set by using information we have gleaned from the test set, such as the fact that a particular term is a good predictor in the test set (even though this is not the case in the training set). A more subtle example of using knowledge about the test set is to try a large number of values of a parameter (e.g., the number of selected features) and select the value that is best for the test set. As a rule, accuracy on new data – the type of data we will encounter when we use the classifier in an application – will be much lower than accuracy on a test set that the classifier has been tuned for. We discussed the same problem in ad hoc retrieval in Section 8.1 (page 153).

DEVELOPMENT SET

HELD-OUT DATA

In a clean statistical text classification experiment, you should never run any program on or even look at the test set while developing a text classification system. Instead, set aside a *development set* for testing while you develop your method. When such a set serves the primary purpose of finding a good value for a parameter, for example, the number of selected features, then it is also called *held-out data*. Train the classifier on the rest of the training set with different parameter values, and then select the value that gives best results on the held-out part of the training set. Ideally, at the very end, when all parameters have been set and the method is fully specified, you run one final experiment on the test set and publish the results. Because no informa-